LAB 1 :

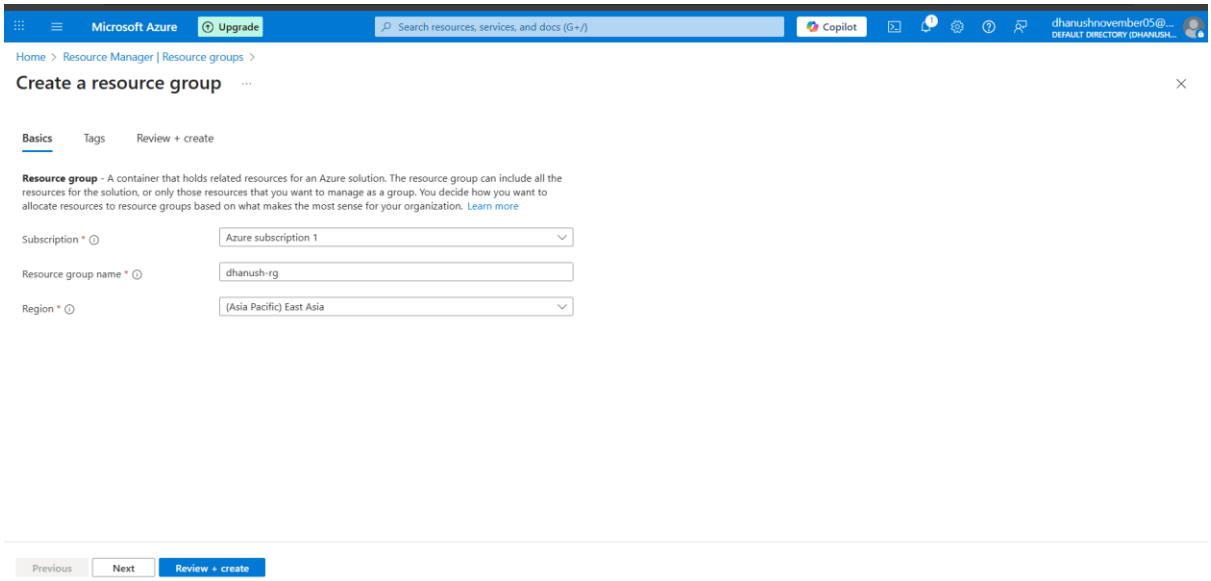
PART A : Create Linux Virtual Machine

STEP 1: Sign in to the Azure Portal → https://portal.azure.com.

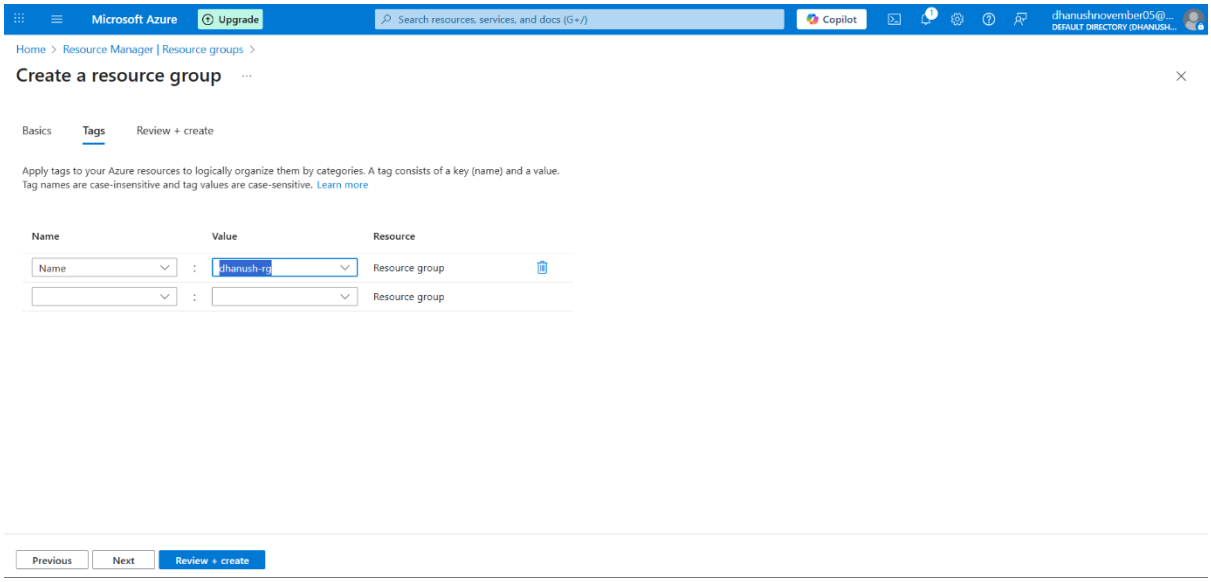
In the left menu, click Resource groups → Create

STEP2 : Resource group name: dhanush-rg

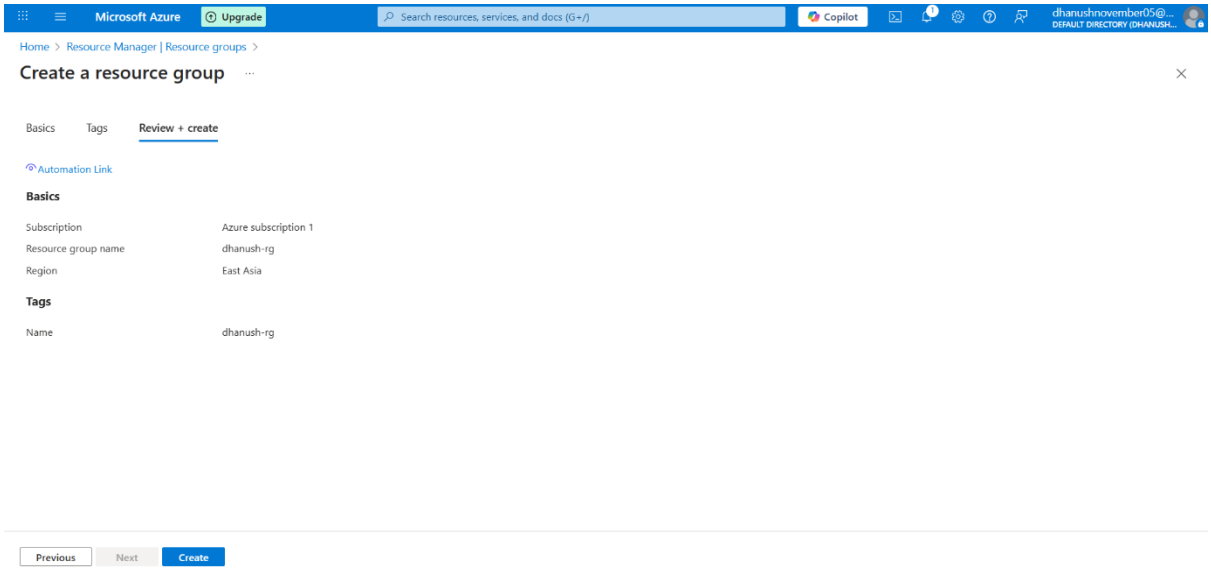
STEP3 : Region: Choose the nearest region

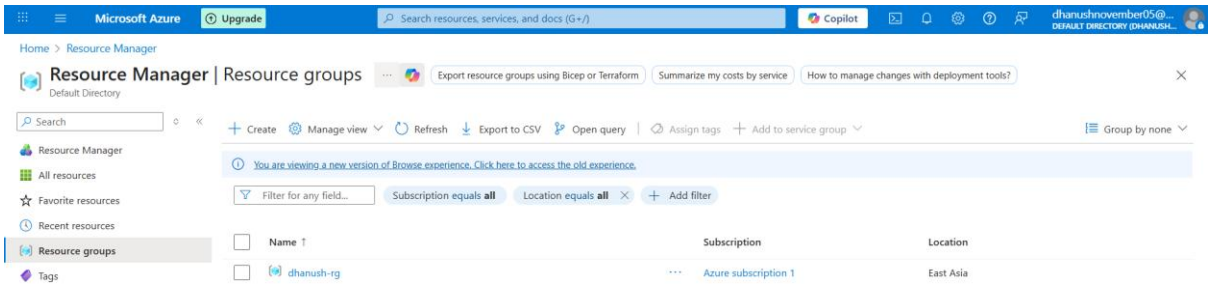


tags

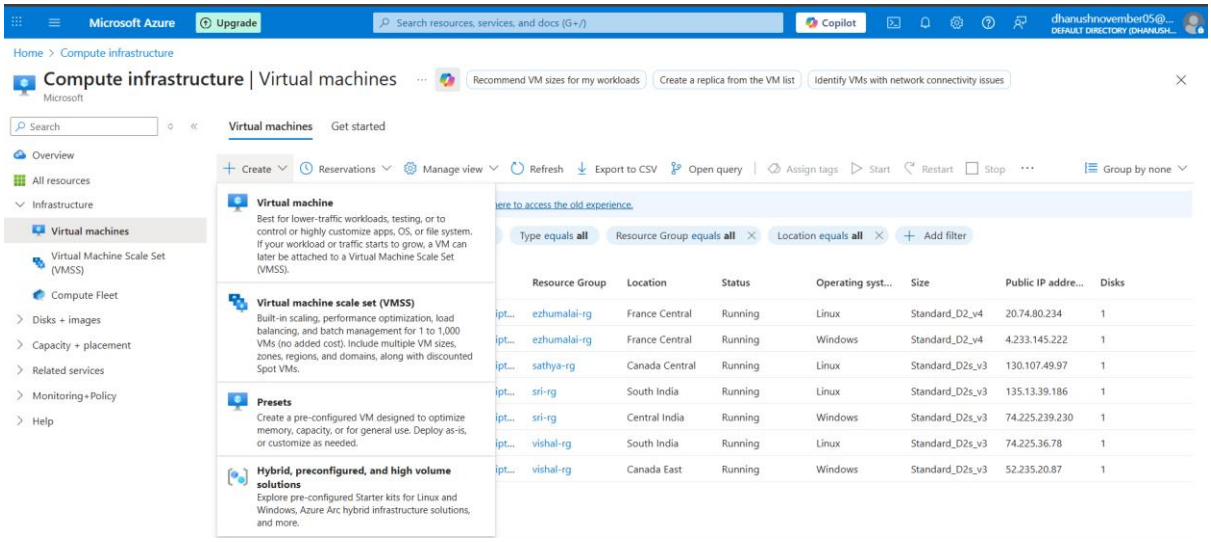


STEP4 : Click Review + Create → Create

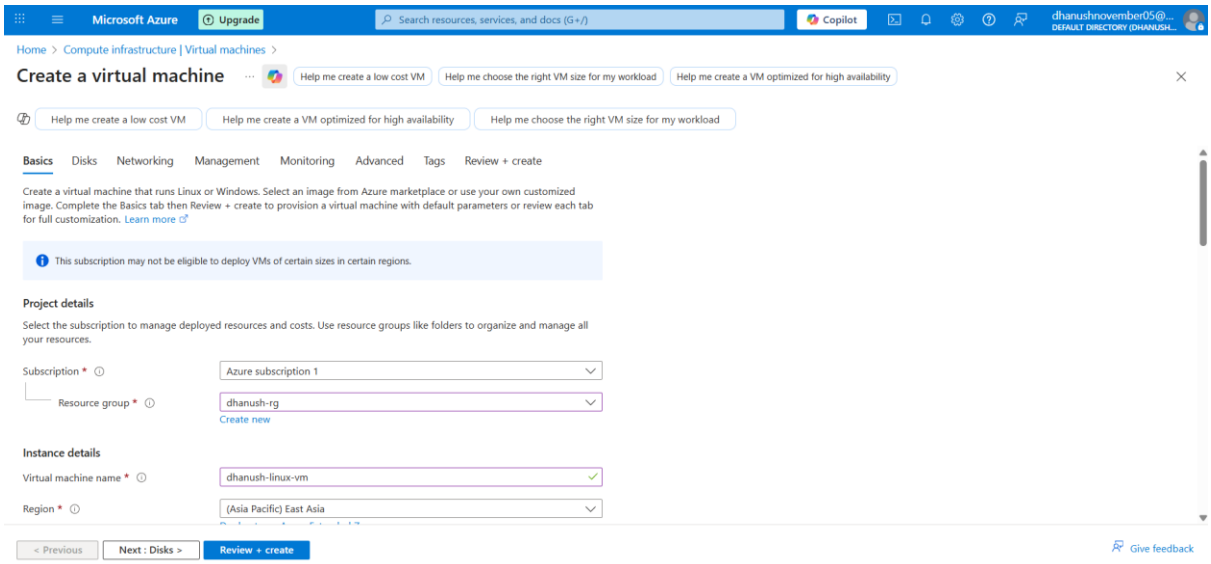




In the search bar, type Virtual machines → Create → Azure Virtual Machine

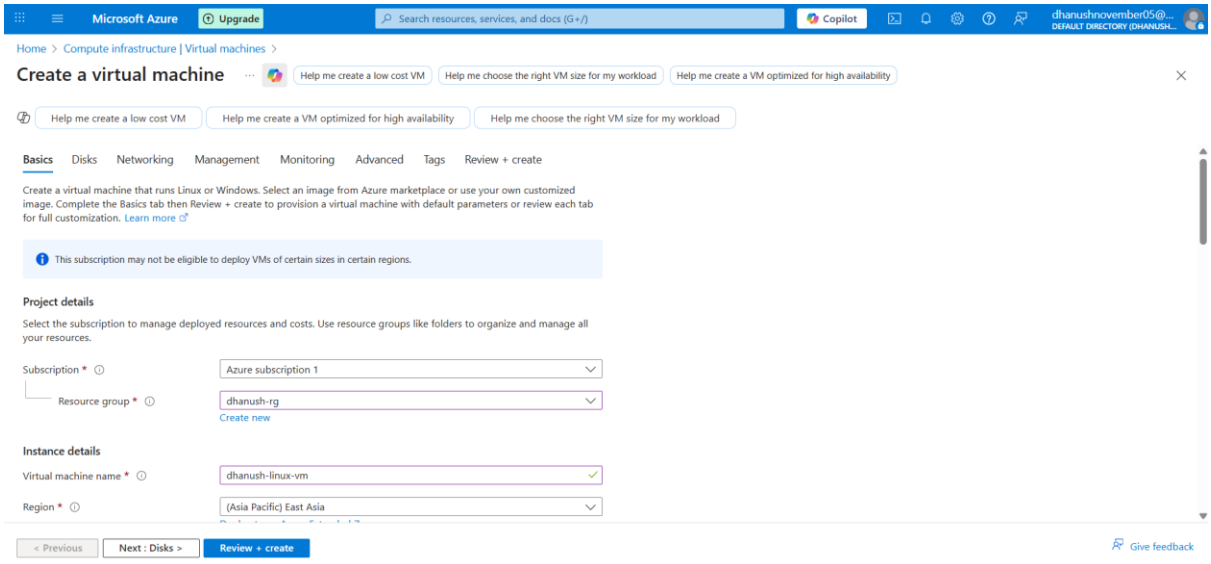


STEP5 : Resource group: <studentname>-rg

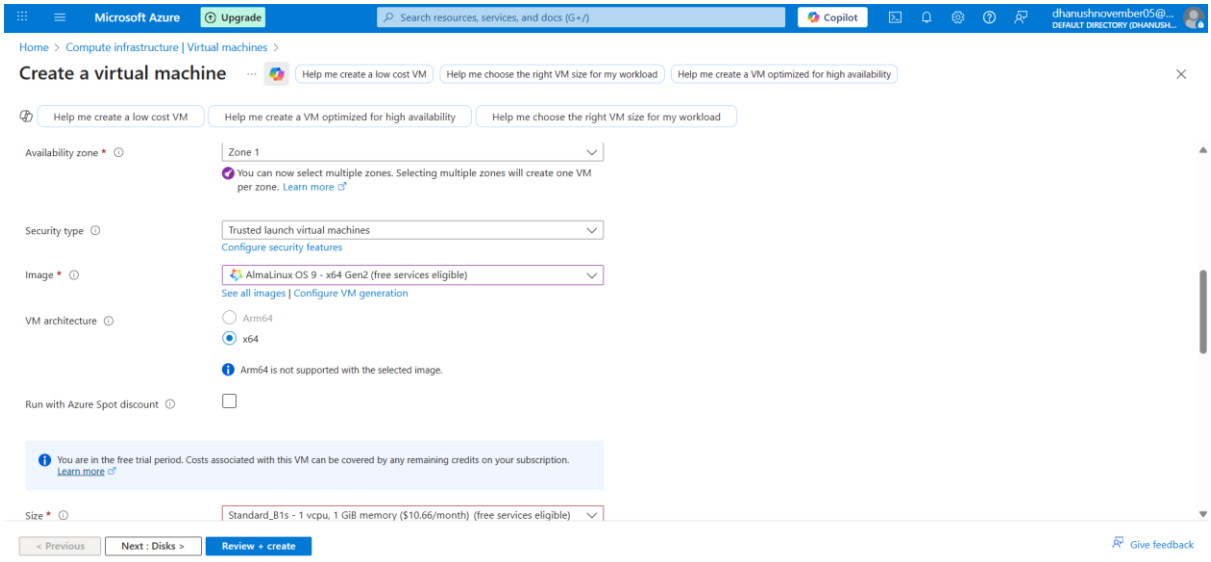


STEP6 : VM name: <studentname>-linux-vm

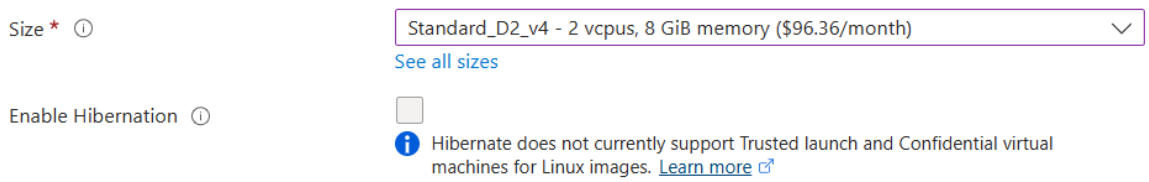
STEP7 : Region: Same as RG



STEP8 : Image: *AlmaLinux OS 9 – x64 Gen2* (or Ubuntu if AlmaLinux not available)



STEP9 : Size: Standard_B1s (free tier eligible) (OR) Standard_D2s_v3



STEP10 : Authentication type: choose Password (OR) SSH public key

Authentication type: Password

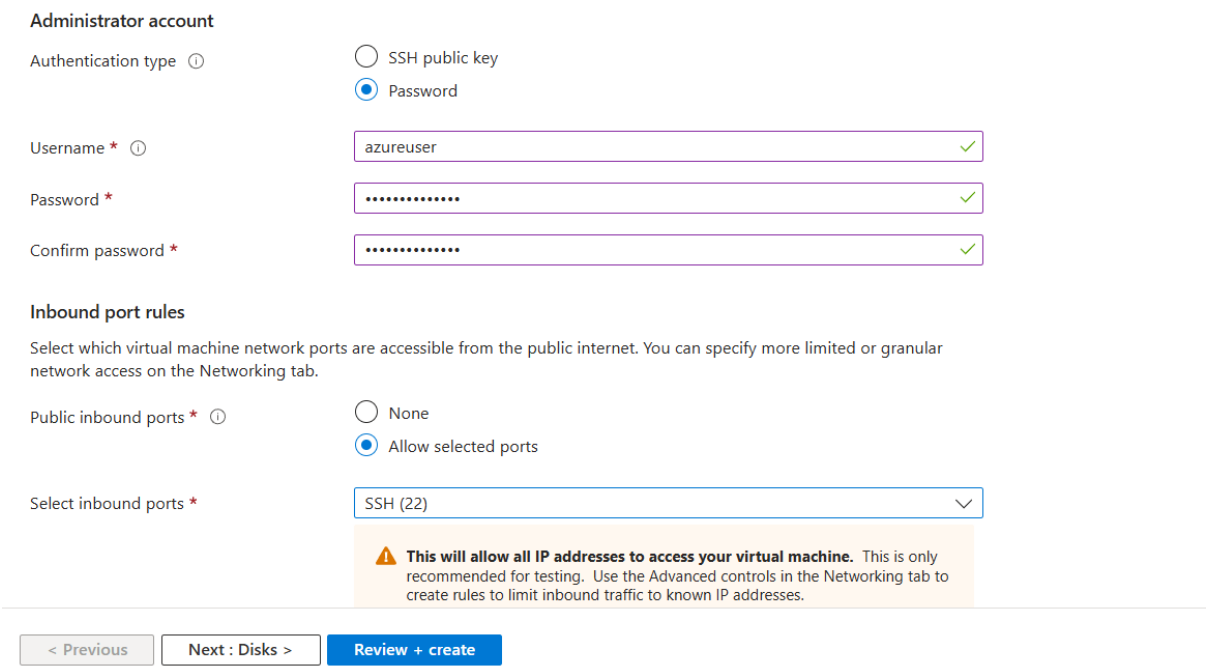
- Username: azureuser
- Password: <choose-strong-password>

(or)

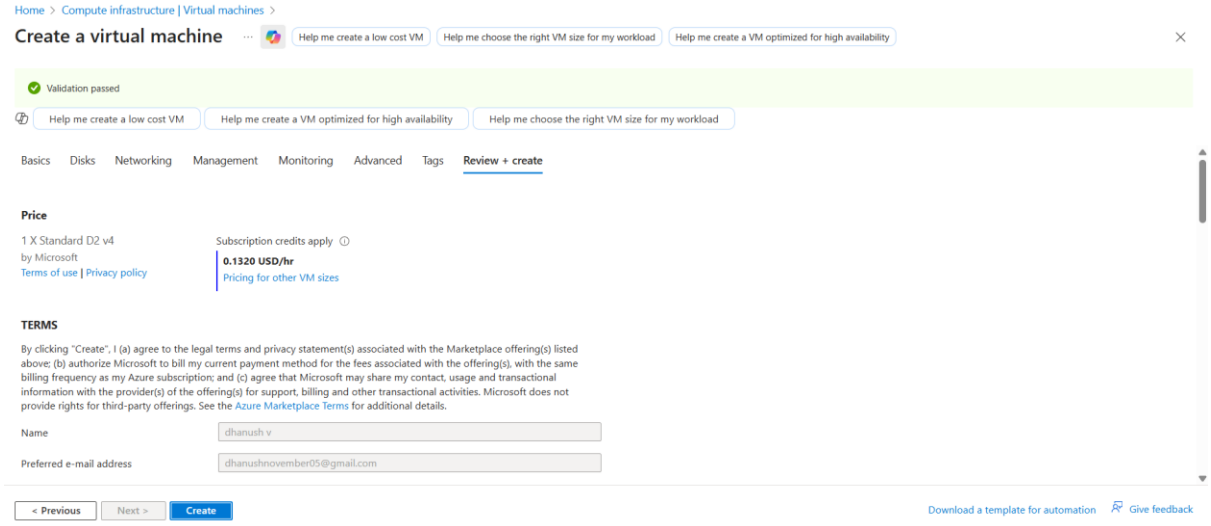
Authentication type: SSH public key

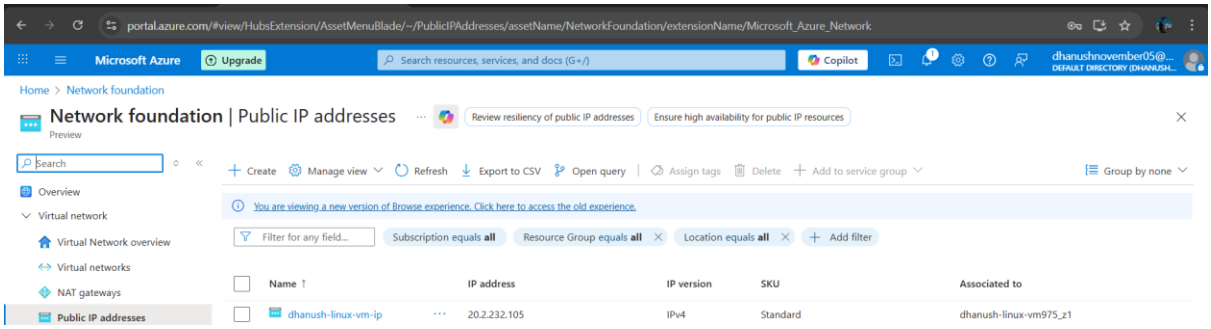
- Generate new key pair → Key name: <studentname>-key

STEP 11: Inbound ports: Allow SSH (22)



STEP12 : Click Review + Create → Create





STEP13: Linux VM is created. You can connect using:

ssh azureuser@<public-ip>

ssh -i <studentname>-key.pem azureuser@<public-ip>

```
C:\Users\DHANUSH>ssh azureuser@20.2.232.105
The authenticity of host '20.2.232.105 (20.2.232.105)' can't be established.
ED25519 key fingerprint is SHA256:SAzOttMgXq9akBfuaYQ4JRmC74gA94cN21cnqJ9q4QQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.2.232.105' (ED25519) to the list of known hosts.
azureuser@20.2.232.105's password:
[azureuser@dhanush-linux-vm ~]$
```

```
[azureuser@dhanush-linux-vm ~]$ sudo su -
[root@dhanush-linux-vm ~]# whoami
root
[root@dhanush-linux-vm ~]# hostname
dhanush-linux-vm
[root@dhanush-linux-vm ~]# yum install net-tools wget zip -y
AlmaLinux 9 - AppStream                                          9.3 MB/s |  12 MB     00:01
AlmaLinux 9 - BaseOS                                             6.0 MB/s | 7.4 MB     00:01
AlmaLinux 9 - Extras                                              17 kB/s |  20 kB     00:01
Last metadata expiration check: 0:00:01 ago on Tue 23 Dec 2025 04:01:27 PM UTC.
Dependencies resolved.
================================================================================================================================================
 Package                      Architecture                 Version                                  Repository                            Size
================================================================================================================================================
Installing:
 net-tools                    x86_64                       2.0-0.64.20160912git.el9                 baseos                               294 k
 wget                         x86_64                       1.21.1-8.el9_4                           appstream                            768 k
 zip                          x86_64                       3.0-35.el9                               baseos                               263 k
Installing dependencies:
 unzip                        x86_64                       6.0-59.el9                               baseos                               179 k

Transaction Summary
================================================================================================================================================
Install  4 Packages

Total download size: 1.5 M
Installed size: 5.0 M
Downloading Packages:
(1/4): unzip-6.0-59.el9.x86_64.rpm                               8.0 MB/s | 179 kB     00:00
(2/4): net-tools-2.0-0.64.20160912git.el9.x86_64.rpm             5.5 MB/s | 294 kB     00:00
(3/4): zip-3.0-35.el9.x86_64.rpm                                 6.2 MB/s | 263 kB     00:00
```

```
[root@dhanush-linux-vm ~]# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.4  netmask 255.255.255.0  broadcast 10.0.0.255
        inet6 fe80::222:48ff:feeb:bce3  prefixlen 64  scopeid 0x20<link>
        ether 00:22:48:eb:bc:e3  txqueuelen 1000  (Ethernet)
        RX packets 5138  bytes 23691485 (22.5 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 4628  bytes 1005043 (981.4 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth1: flags=6211<UP,BROADCAST,RUNNING,SLAVE,MULTICAST>  mtu 1500
        ether 00:22:48:eb:bc:e3  txqueuelen 1000  (Ethernet)
        RX packets 15823  bytes 22725172 (21.6 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 4800  bytes 1016517 (992.6 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

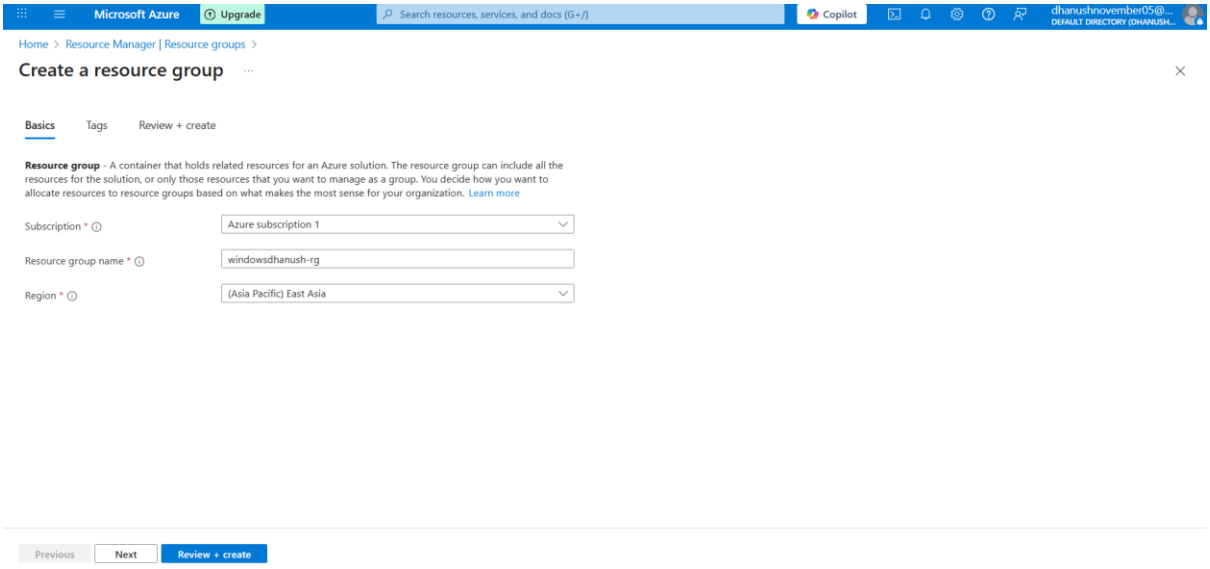
lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 6  bytes 416 (416.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 6  bytes 416 (416.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[root@dhanush-linux-vm ~]# curl ifconfig.me
20.2.232.105[root@dhanush-linux-vm ~]#
```

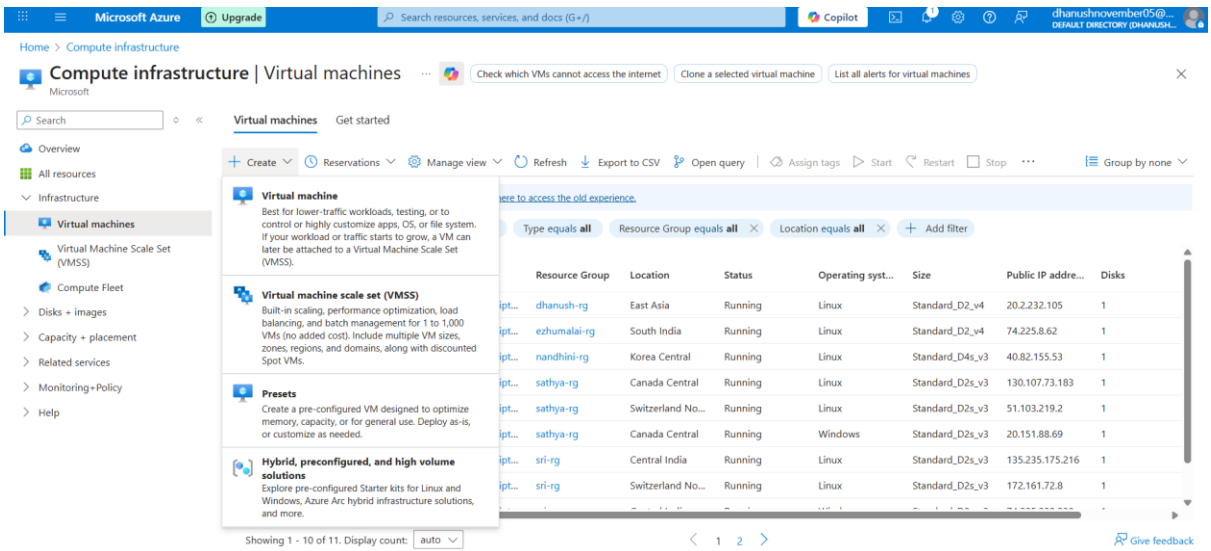
PART B : Create Windows Virtual Machine

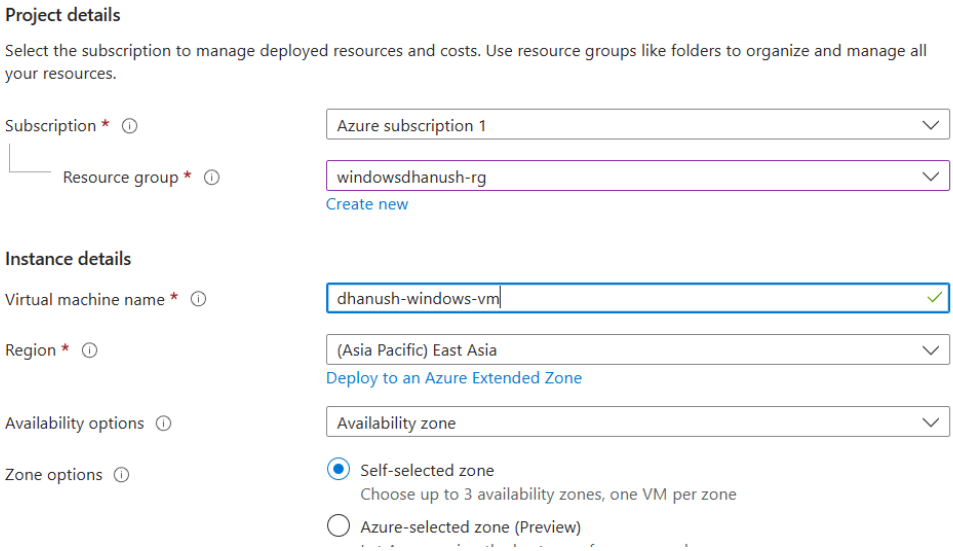
In Azure Portal, go to Virtual machines → Create → Azure Virtual Machine

- **Resource group:** windowsdhanush-rg

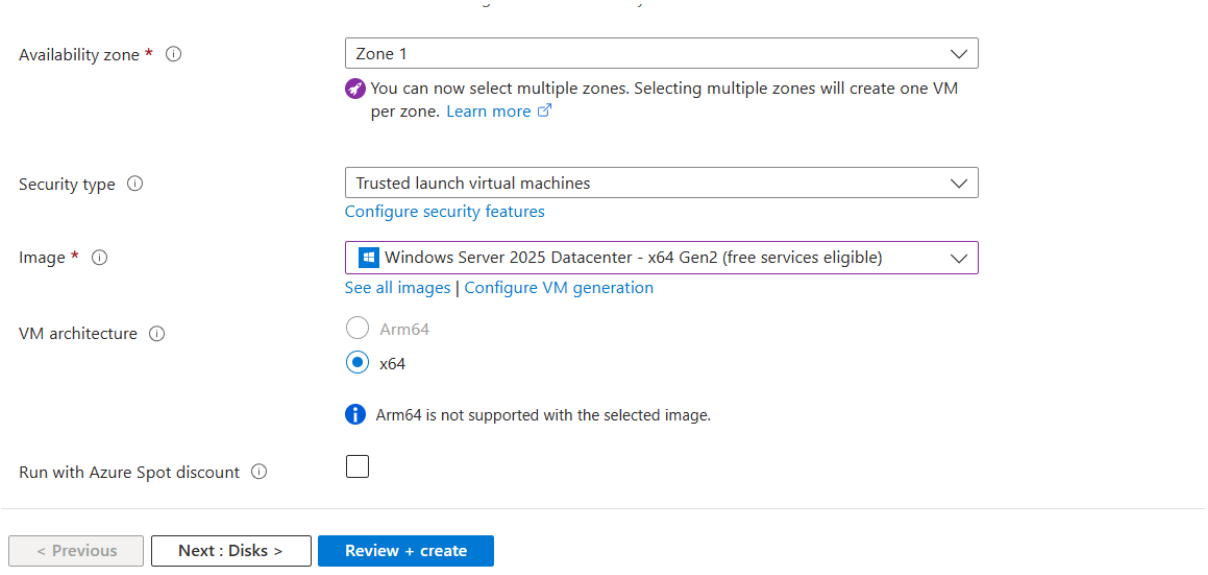


- **VM name:** dhanush-windows-vm
- **Region:** Same as Linux VM



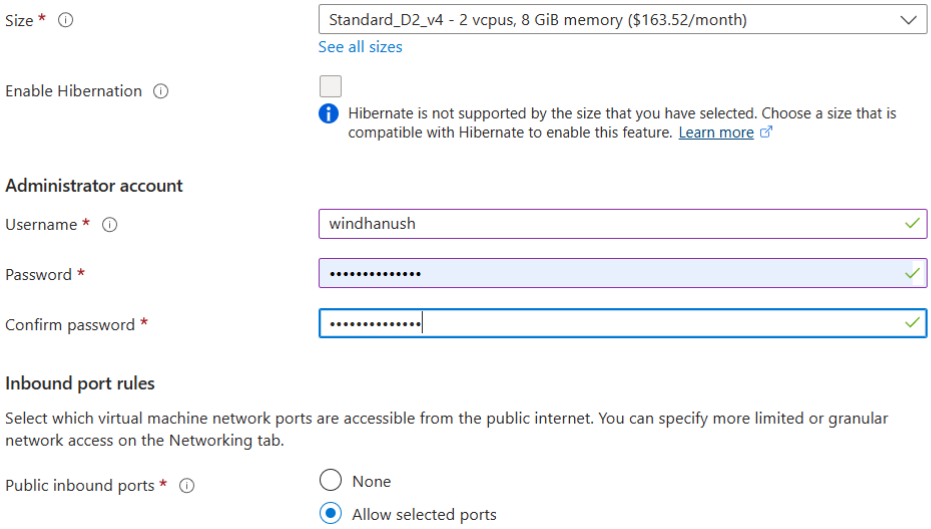


- **Image:** *Windows Server 2022 Datacenter – x64 Gen2 (OR) Windows Server 2025 Datacenter – x64 Gen2*

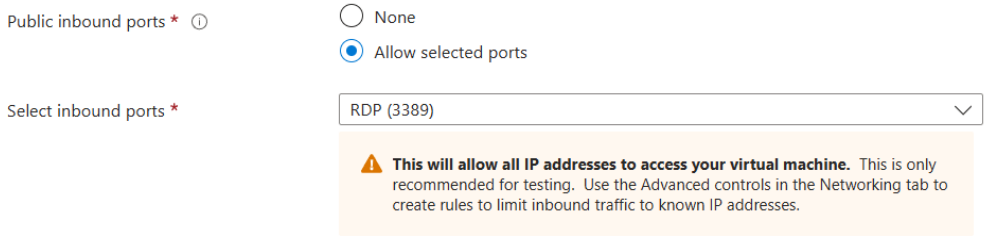


- **Size:** Standard_B2s (or free tier eligible if available) (OR) Standard_D2s_v3
- **Authentication type:** Password
    - Username: azureuser (or) windhanush

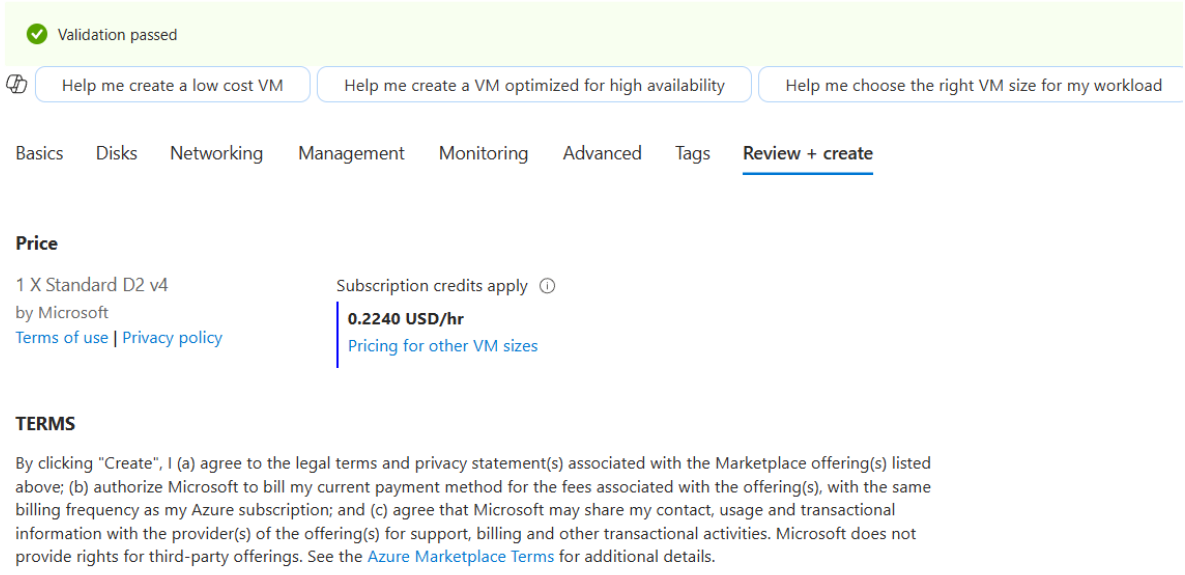
- Password: <choose-strong-password>



- **Inbound ports:** Allow RDP (3389)

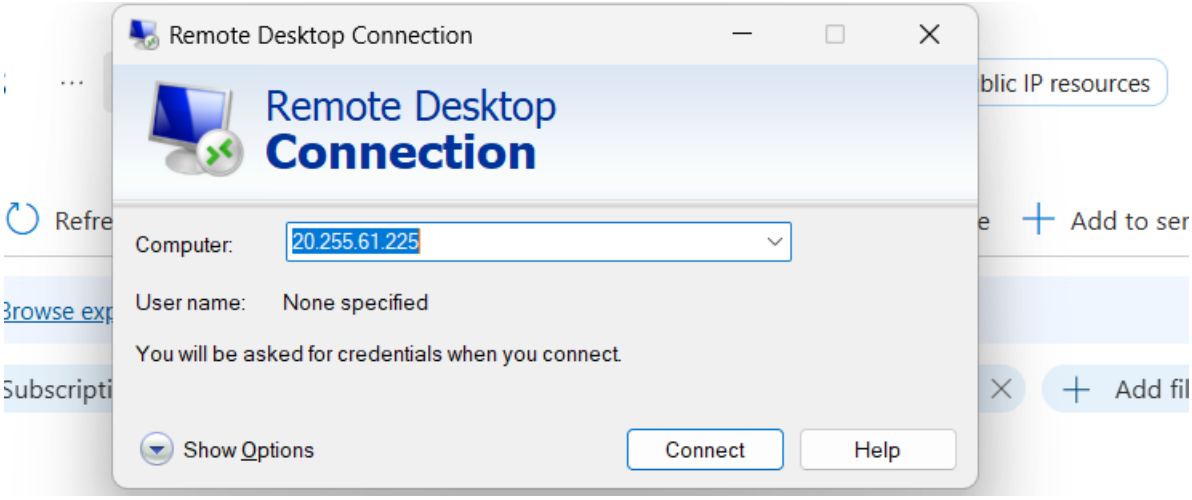


- Click **Review + Create → Create**

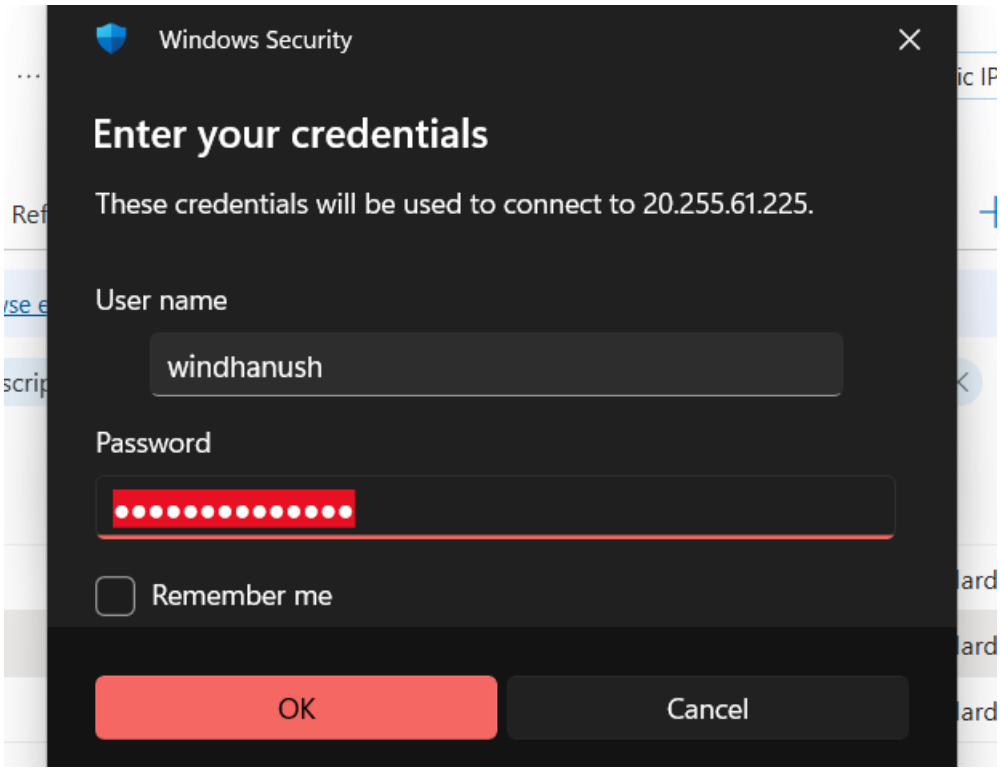


Windows VM is created. You can connect using Remote Desktop (RDP):

- Open **Remote Desktop Connection** (mstsc on Windows)



- Enter VM's **Public IP**
- Login with username azureuser and your password.



Part 1 – Create Linux Virtual Machine

1. Sign in to the **Azure Portal** → https://portal.azure.com.
2. In the left menu, click **Resource groups** → **Create**
    - Resource group name: <studentname>-rg
    - Region: Choose the nearest region
    - Click **Review + Create → Create**
3. In the search bar, type **Virtual machines** → **Create → Azure Virtual Machine**
    - **Resource group:** <studentname>-rg
    - **VM name:** <studentname>-linux-vm
    - **Region:** Same as RG
    - **Image:** *AlmaLinux OS 9 – x64 Gen2* (or Ubuntu if AlmaLinux not available)
    - **Size:** Standard_B1s (free tier eligible) (OR) Standard_D2s_v3
    - **Authentication type:** choose Password (OR) SSH public key
    - **Authentication type:** Password
        - Username: azureuser
        - Password: <choose-strong-password>
    - **Authentication type:** SSH public key
        - Generate new key pair → Key name: <studentname>-key

- **Inbound ports:** Allow SSH (22)
- Click **Review + Create → Create**
- Download the **private key (.pem)** if prompted.

✅ Linux VM is created. You can connect using:

ssh azureuser@<public-ip>

ssh -i <studentname>-key.pem azureuser@<public-ip>

Verification

- **Linux VM** → run:
- whoami
- hostname
- yum install net-tools wget zip -y
- ifconfig

curl ifconfig.me

Part 2 – Create Windows Virtual Machine

1. In Azure Portal, go to **Virtual machines** → **Create** → **Azure Virtual Machine**
   - **Resource group:** <studentname>-rg
   - **VM name:** <studentname>-windows-vm

- **Region:** Same as Linux VM
- **Image:** *Windows Server 2022 Datacenter – x64 Gen2 (OR) Windows Server 2025 Datacenter – x64 Gen2*
- **Size:** Standard_B2s (or free tier eligible if available) (OR) Standard_D2s_v3
- **Authentication type:** Password
  - Username: azureuser
  - Password: <choose-strong-password>
- **Inbound ports:** Allow RDP (3389)
- Click **Review + Create → Create**

✅ Windows VM is created. You can connect using **Remote Desktop (RDP)**:

- Open **Remote Desktop Connection** (mstsc on Windows)
- Enter VM's **Public IP**
- Login with username azureuser and your password.

- **Windows VM** → open **PowerShell** and run:
- whoami ; hostname
- ipconfig

curl ifconfig.me